The *general correlation problem* considers sums of the form.

$$S = \sum a(n)b(n).$$

For example, taking $a(n) = 1$ and $b(n) = n^{-it}$ is relevant for $\zeta(\frac{1}{2} + it)$.

We assume that they are supported in $[N, 2N]$.

We also assume that they are essentially O(1) in mean square ("Ramanujan on average").

We might also normalize each coefficient sequence to have mean zero, by subtracting the average.

Then, by Cauchy–Schwarz,

$$S \ll N^{1+\varepsilon}.$$

The sequences are said to be *independent* if $S \ll N^{1-\delta}$.

These problems are important in various applications:

- subconvexity problems
- shifted convolution problems
- Dirichlet divisor problem, Gauss circle problem, critical zeros of L-functions, ...

We will discuss mainly the *delta symbol approach* towards these correlation problems. In that, we first consider the space $\mathcal{B}$ of all functions $[N, 2N] \cap \mathbb{Z} \to \mathbb{C}$. This is a vector space of dimension N with hermitian inner product $\langle , \rangle$. Note that $a = \{a(n)\}, b = \{b(n)\} \in \mathcal{B}$, and

$$S = \langle a, \bar{b} \rangle. \tag{1}$$

Suppose $\{\psi_v(n) : v \in \mathcal{B}\}$ is an orthonormal basis of the space $\mathcal{B}$. Then we get an orthonormal matrix

$$\psi_v(n)\overline{\psi_v}(m) =: \delta(n, m) = \begin{cases} 1 & \text{if } n = m, \\ 0 & \text{otherwise.} \end{cases}$$

We can use this basis to expand (1) as a sum of products

$$\langle a, \psi_v \rangle \langle \psi_v, b \rangle.$$

We can often use Poisson, Voronoi, etc., to transform

$$\langle a, \psi_v \rangle = \sum_{n \asymp N} a(n)\overline{\psi_v(n)} \approx \sum_{n \asymp N^*} a^*(n)\overline{\psi_v^*(n)},$$

and similarly for b. Then, applying Cauchy–Schwarz to get rid of (say) a, we obtain

$$S \ll \sqrt{N^*} \left(\sum_{n \asymp N^*} \left| \sum_{v \in \mathcal{B}} \psi_v^*(n) \sum_{m \asymp M^\dagger} b^\dagger(m) \psi_v^\dagger(m) \right|^2 \right)^{1/2}.$$

Opening the absolute value squared, we then use summation formulas to estimate the sums

$$\sum_{n \asymp N^*} \psi_v^*(n) \overline{\psi_{v'}^*(n)}.$$

The classical form of the circle method is based on the trigonometric functions, or harmonics of the circle group.

The skeleton for the GL(1) delta symbol formula is

$$\frac{1}{N}\sum_{q<\sqrt{N}}\sum_{a(q)^*} e_q(an)\overline{e_q(am)}.$$

Here we basically take $v=(a,q)$.

DFI delta method (Duke, Friedlander, Iwaniec): for a large real number $L\geq 1$ and $n\in\mathbb{Z}\cap[-2L,2L]$,

$$\delta(n)=\frac{1}{Q}\sum_{1\leq q\leq Q}\frac{1}{q}\sum_{a(q)^*} e_q(an)\int_{\mathbb{R}} g(q,x)e\left(\frac{nx}{qQ}\right)\,dx,$$

where $Q=2L^{1/2}$ and $e(z)=e^{2\pi iz}$. The function g is not given explicitly, but we only need a few properties in our analysis. This leads to

$$\delta(n)=\frac{1}{Q}\sum_{1\leq q\leq Q}\frac{1}{q}\sum_{a(q)^*} e_q(an)\int_{\mathbb{R}} W\left(\frac{x}{Q^{\varepsilon}}\right)g(q,x)e\left(\frac{nx}{qQ}\right)\,dx,$$

where W is mild.

Let's now give some rough basics on modular forms. A holomorphic function $f:\mathbb{H}\to\mathbb{C}$ is a modular form of weight k for $\mathrm{SL}_2(\mathbb{Z})$ if it satisfies the automorphy condition $f(\gamma z)=(cz+d)^k f(z)$ for all $\gamma\in\mathrm{SL}_2(\mathbb{Z})$ and satisfies a growth condition. It then admits a Fourier expansion $\sum_{n\geq 0}a(n)q^n$, and is a cusp form if $a_0=0$.

Voronoi summation formula says that if $\gcd(a,c)=1$, then

$$\sum a(n)e_c(an)W(n)=\frac{1}{c}\sum_n a(n)e_c(-\bar{a}n)\int(\text{Bessel transform of W}).$$

Dirichlet divisor problem: taking $a(n)=1$ and $b(n)=d(n)$, the sum S is as in the Dirichlet divisor problem. Hyperbola method.

$$S=X\log X+X(2\gamma-1)+\mathrm{O}(X^{1-\theta})$$

Hardy: $\theta\geq 1/4$. Huxley (2003): $\theta=131/416$

Now consider

$$A(x)=\sum_{n\leq X}a(n),\mathcal{A}(s)=\sum_{n=1}^{\infty}\frac{a(n)}{n^s},$$

degree d L-function (satisfying some properties, e.g., Selberg class).

Friedlander–Iwaniec: $A(X)=R(X)+\Delta(X)$, where $R(X)$ is given explicitly as the residue $X\,\mathrm{res}_{s=1}\frac{\mathcal{A}(s)}{s}$ and

$$\Delta(X)\ll X^{\frac{d-1}{d+1}+\varepsilon}.$$

Friedlander–Iwaniec conjectured that

$$\Delta(X)\ll X^{\frac{d-1}{2d}+\varepsilon}.$$

For $d=2$, it remains a great open problem to improve the bound (for a cusp form) $\Delta(X)\ll X^{1/3}$.

We will focus mainly on the case $a(n)=\lambda_f(n)$ and $b(n)=\lambda_g(n)$, where $\lambda_f(n)$ and $\lambda_g(n)$ are normalized Fourier coefficients (i.e., $\lambda_f(1)=1$) of Hecke cusp forms (Maass or holomorphic). Corresponding L-function:

$$L(s,f\otimes g)=\zeta(2s)\sum_{n=1}^{\infty}\frac{\lambda_f(n)\lambda_g(n)}{n^s}.$$

For this case, we have

$$A(x) = C_f 1_{f=\bar{g}} + \mathrm{O}(X^{3/5+\varepsilon}),$$

where for $f = \bar{g}$, we have

$$C_f = \frac{L(1, \mathrm{sym}^2 f)}{\zeta(2)}.$$

We want to discuss the problem first when $f = \bar{g}$, of degree 4.

Ivic first proved that $\Delta(X) = \mathrm{O}_f(X^{3/8+\varepsilon})$, and Lau-Lu–Wu proved that $\Delta(X, f) = \Omega_f(X^{3/8})$.

Huang first considered the problem when $f = \bar{g}$, and showed that

$$\Delta(X) \ll X^{\frac{3}{5} - \frac{1}{560} + \varepsilon}.$$

Later, Pal considered the second moment of degree three L-functions and got

$$\Delta(X) \ll X^{\frac{3}{5} - \frac{6}{1085} + \varepsilon}.$$

Huang improved the previous bound and got

$$\Delta(X) \ll X^{\frac{3}{5} - \frac{3}{305} + \varepsilon}.$$

Kumar–Sharma got

$$\Delta(X) \ll X^{\frac{3}{5} - \frac{3}{205} + \varepsilon}.$$

Recently, Dasgupta, Leung and Young considered the second moment of degree three L-functions and got

$$\Delta(X) \ll X^{\frac{3}{5} - \frac{1}{35} + \varepsilon}.$$

Huang considered

$$\lambda_{1 \boxplus \mathrm{sym}^2 f}(n) = \lambda_{f \otimes \bar{f}}(n) = \lambda_{f \otimes f}(n) = \sum_{\ell^2 m = n} \lambda_f(m)^2.$$

Instead of $\mathrm{sym}^2 f$, Huang considered a Hecke–Maass cusp form Φ for $\mathrm{SL}_3(\mathbb{Z})$ and considered the sum

$$\mathcal{A}(X, 1 \boxplus \Phi) = \sum_{1 \leq n \leq X} \lambda_{1 \boxplus \Phi}(n),$$

where $\lambda_{1 \boxplus \Phi}(n) = \sum_{\ell m = n} A_\Phi(1, m)$. (Or smoothed variant.)

Using the inverse Mellin transforms, he reduced the problem that of estimating a first integral moment of L-functions:

$$\int_T^{2T} L(\tfrac{1}{2} + it, 1 \boxplus \Phi) X^{it}\, dt, \quad T = X^{\frac{2}{5} + \delta}, \quad \delta > 0.$$

Then he reduced to estimating

$$\sum_{n \geq 1} \lambda_{1 \boxplus \Phi}(n) e(-4(nX))^{1/4} V\left(\frac{n}{N}\right).$$

Using the delta method, he proved that

$$\sum_{n \geq 1} \lambda_{1 \boxplus \Phi}(n) W\left(\frac{n}{N}\right) = L(1, \Phi) X + \mathrm{O}(X^{\frac{3}{5} - \delta + \varepsilon}).$$

To get the required result, at last he used

$$\lambda_f(n)^2 = \sum_{\ell^2 m = n}\sum \cdots .$$

In an ongoing project (joint with Mallesham, Munshi and Singh), we will show that:

Theorem 1. *Let f and g be Hecke cusp forms for $\mathrm{SL}_2(\mathbb{Z})$. Then there exists $\delta > 0$ such that*

$$\sum_{n \le X} \lambda_f(n)\lambda_g(n) = C_f 1_{f=\bar{g}} X + \mathrm{O}(X^{\frac{3}{5}-\delta+\varepsilon}).$$

We now sketch the proof. By dyadic subdivision, reduce to summing over $n \asymp X$. Smooth the sum out using W supported on $[1-X^{\sigma-1}, 2+\Xi^{\sigma-1}]$, taking the value 1 on $[1,2]$; this gives S up to accuracy $\mathrm{O}(X^\sigma)$. Using Mellin transform and stationary phase method. Focus on

$$\sum_{n \asymp N} \lambda_f(n)\lambda_g(n) e\left(-4(nX)^{1/4}\right).$$

Here we are avoiding the method described in Friedlander–Iwaniec's paper, since we don't want to use the pointwise Ramanujan conjecture.

We will consider a more general correlation problem:

$$S := \sum_{n \asymp N} \lambda_f(n)\lambda_g(n) e(f(n)),$$

with $f(n) \asymp N^\theta$.

We are working with general phase functions $f(n)$ of size N^θ (e.g., $f(n) = \alpha n^\beta$ for some $0 \ne \alpha \in \mathbb{R}$), and will show that $\theta < \frac{7}{10}$. For the Rankin–Selberg problem, we have the special phase function $f(n) = -4(nX)^{1/4}$ with $n \asymp N$ and $X \asymp N^{5/3}$. Hence the amplitude here is of size $(NX)^{1/4} = N^{2/3}$, i.e., $\theta = 2/3$. This falls in the range covered by our result as $2/3 < 7/10$. (Note that the Weyl range would be $3/4$, which is far away from $7/10$.)

We use the DFI delta symbol. This leads to

$$S \approx \frac{1}{Q^2} \int_{\mathbb{R}} \sum_{q \asymp Q} \sum_{a(q^*)} \sum_{n \asymp N} \lambda_f(n) e_q(an) e\left(\frac{nu}{qQ}\right) \sum_{m \asymp N} \lambda_g(m) e_q(-am) e(f(m)) e(f(m)) e\left(-\frac{mu}{qQ}\right) du.$$

Trivial estimation at this stage gives $N^{2+\varepsilon}$, so we need to say $N^{1+\cdots}$.

Voronoi gives

$$\sum_n \cdots \rightsquigarrow \frac{N}{q} \sum_{n \asymp \frac{(QK)^2}{N} = K} \lambda_f(n) e_q(-\bar{a}n) \mathcal{I}_1,$$

saving $\sqrt{N/K}$, and

$$\sum_m \cdots \rightsquigarrow \frac{N}{q} \sum_{m \asymp \frac{(QN^\theta)^2}{N} = \frac{N^{2\theta}}{K}} \lambda_g(m) e_q(\bar{a}m) \mathcal{I}_2,$$

saving $\sqrt{NK}/N^\theta$.

In total, we obtain

$$S \approx \sum_{q \asymp Q} \sum_{n \asymp K} \lambda_f(n) \sum_{m \asymp \frac{N^{2\theta}}{K}} \lambda_g(m) C_q(n-m) \mathcal{I}(m,n,q),$$

where $\mathcal{I}(m,n,q)$ is the whole integral. Savings from the integral is $\sqrt{K}$. Savings from the a-sum is $Q = \sqrt{N/K}$. Total savings is

$$\sqrt{\frac{N}{K}} \cdot \frac{\sqrt{NK}}{N^\theta} \cdot \sqrt{K} \cdot \sqrt{\frac{N}{K}} = N \cdot N^{\frac{1}{2}-\theta}.$$

We need to do more. Apply Cauchy–Schwarz, focus on

$$\sum_{q \asymp Q} \sum_{m \asymp \frac{N^{2\theta}}{K}} \left| \sum_{n \asymp K} \lambda_f(n) C_q(n-m) \mathcal{J}(m,n,q) \right|^2,$$

where $\mathcal{J}$ has size K and open the absolute value squared, giving sums over q, m, n_1, n_2. For the diagonal $n_1 = n_2$, we save $\frac{K}{Q} = \frac{K^{3/2}}{\sqrt{N}}$. For the off-diagonal $n_1 \neq n_2$, we focus on the resulting m-sum

$$\sum_{m \asymp \frac{N^{2\theta}}{K}} C_q(n_1 - m) C_q(n_2 - m) \mathcal{J}(m, n_1, q) \mathcal{J}(m, n_2, q).$$

We use Poisson summation on the m-sum. This gives rise to a character sum modulo q, involving Ramanujan sums:

$$\sum_{\beta(q)} C_q(n_1 - \beta) C_q(n_2 - \beta) e_q(m\beta).$$

After opening these and evaluating, we get

$$\asymp q^2 e_q(mn_q) 1_{n_1 \equiv n_2 (q)},$$

which is $\asymp q$ on average. Thus far, for the m-sum, we obtain

$$\frac{N^{2\theta}}{K} \mapsto \frac{N^{2\theta}}{KQ} \cdot \frac{K^{3/2}}{N^{2\theta - 1/2}} \cdot Q \cdot \frac{1}{\sqrt{K}}.$$

Hence savings for the m sum becomes

$$\frac{N^{2\theta-1/2}}{K}.$$

Okay, now, we get some further off-diagonal savings. We're looking at

$$\sum_{m \ll \frac{K^{3/2}}{N^{2\theta-1/2}}} \sum_{q \asymp Q} \sum_{n_1 \equiv n_2 (q)} \lambda_f(n_1) \lambda_f(n_2) e_q(mn_2) \int \cdots.$$

We apply Cauchy–Schwarz, leading to

$$\sum_m \sum_q \sum_{n_1} \left| \sum_{n_2 \equiv n_1 (q)} \cdots \right|^2.$$

Straightforward analysis isn't sufficient. To do better, we improve by using some markers $n \asymp L$ to partition the n_2's into small blocks satisfying

$$|n_2 - \nu| \ll \frac{K}{L}.$$

This works a bit opposite to the amplification method or the mass transfer method – the diagonal is okay but the off-diagonal is not, so we're trying to save more in the off-diagonal. With this modification, the diagonal savings are K/LQ. For off-diagonal, we will mainly consider $\sum_{n_1 \asymp K,\, n_1 \equiv n_2 (q)} \cdots$. Here the effective length for n_1 is K/Q and $\cdots$ oscillates like K/L. Off-diagonal savings becomes

$$\frac{K^{3/2}}{\sqrt{N}\sqrt{K/L}}.$$

Optimal choice for L is obtained by equating diagonal and off-diagonal, giving $L = K^{1/3}$.

To choose the optimal size for K, we solve

$$\frac{K^{3/2}}{\sqrt{N}} = \frac{N^{2\theta-1/2}}{K} \cdot \frac{K^{7/12}}{N^{1/4}},$$

yielding

$$K = N^{\frac{24}{23}\theta - \frac{3}{23}}.$$

Our total savings is thus

$$N \cdot N^{1/2-\theta} \cdot \sqrt{\frac{K^{3/2}}{\sqrt{N}}} = N \cdot N^{\frac{1}{4}-\theta+\frac{3}{4}\left(\frac{24}{23}\theta - \frac{3}{23}\right)}.$$

This savings will be enough for us if this last parenthetical quantity in the exponent is positive. This means $\theta < 7/10$, and as $7/10 > 2/3$, we are done.

References